

MATH6119

Analytical Consultancy Skills
jstall-uos-mathsci

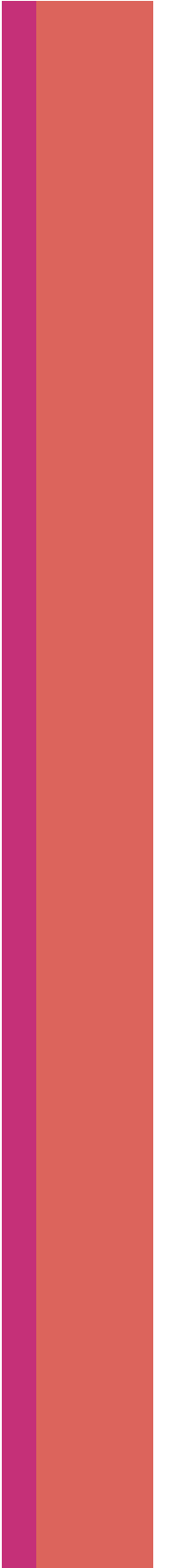


What you'll be doing:

- assigned a problem to tackle in 1 week
- you'll work in groups of 6-7 members
- designing, implement, and report a solution

Deliverables for assessment:

- solution (source code, input files, output, documentation)
- analytical report (document)
- presentation
- evidence of group work (signed meeting minutes)



Expectations

- You will need to spend at least 50% of your time working on this assignment. There is a lot to do and not much time to do it.
- There is no expectation that you work on weekends unless your team agrees to do so.
- Your team will be self-managing. You will internally decide who is responsible for what. Conflicts will need to be dealt with internally as well.
- All team members should be present (either in person or remotely) for all team meetings. This must be evidenced.
- Everyone in a team receives the same grade at the end. You succeed or fail together.

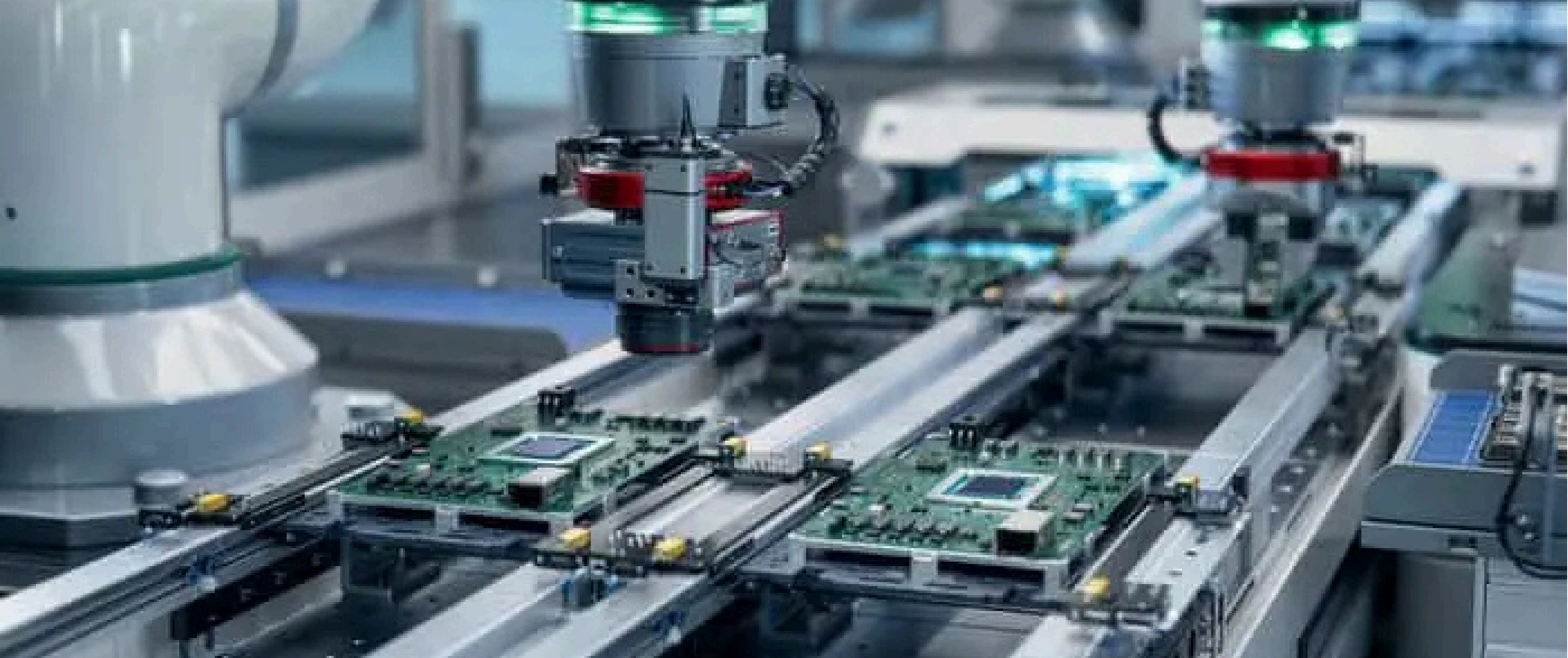
Help during assignment

If you get stuck during the assignment, **don't wait**. Seek help.

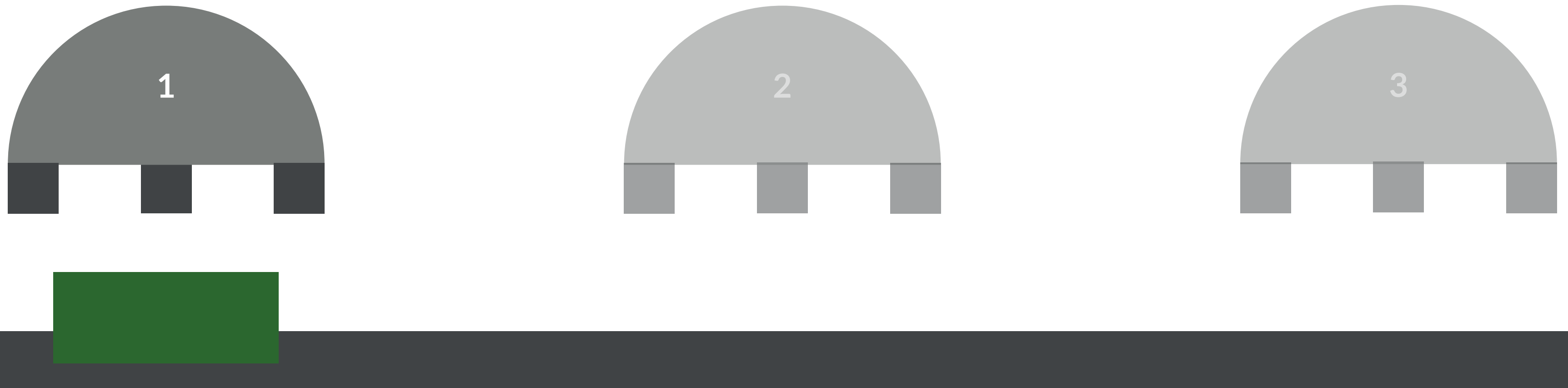
A teaching assistant will be on hand to help you on specific days during the week. Further details of times and locations will be posted on Blackboard.

If you need further clarification on the assignment, email me:

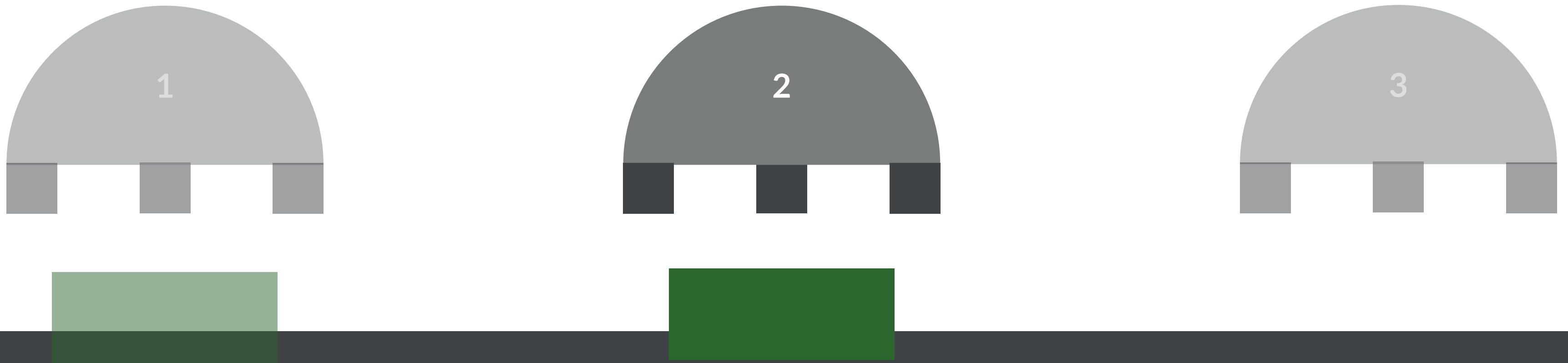
j.e.stallwood@soton.ac.uk



PROBLEM DESCRIPTION

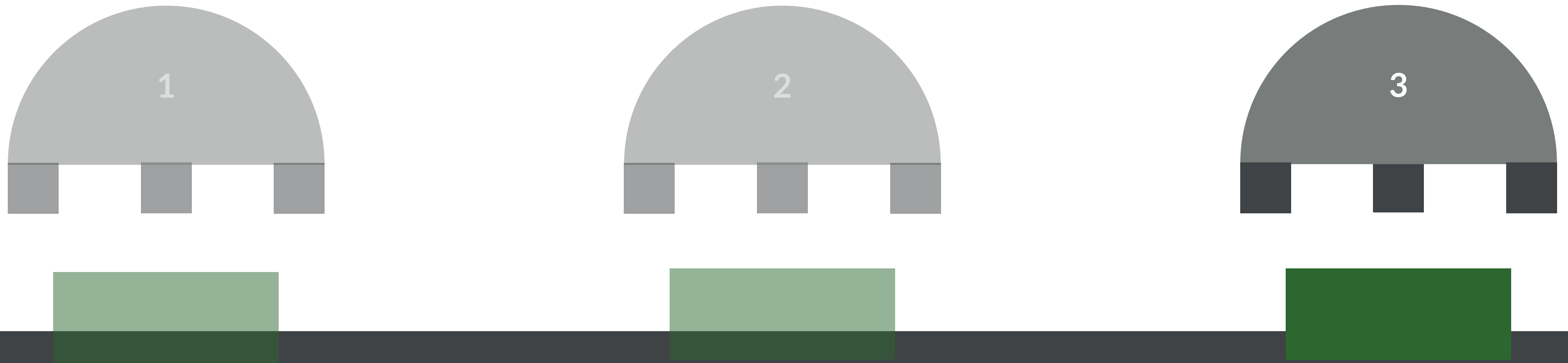


A PCB will come along a conveyor belt. It must be fitted with specific components in certain places. It stops in front of machine 1. Machine 1 picks up and places all the components it is tasked to do (at most 3 components at a time). This process takes a certain amount of time.

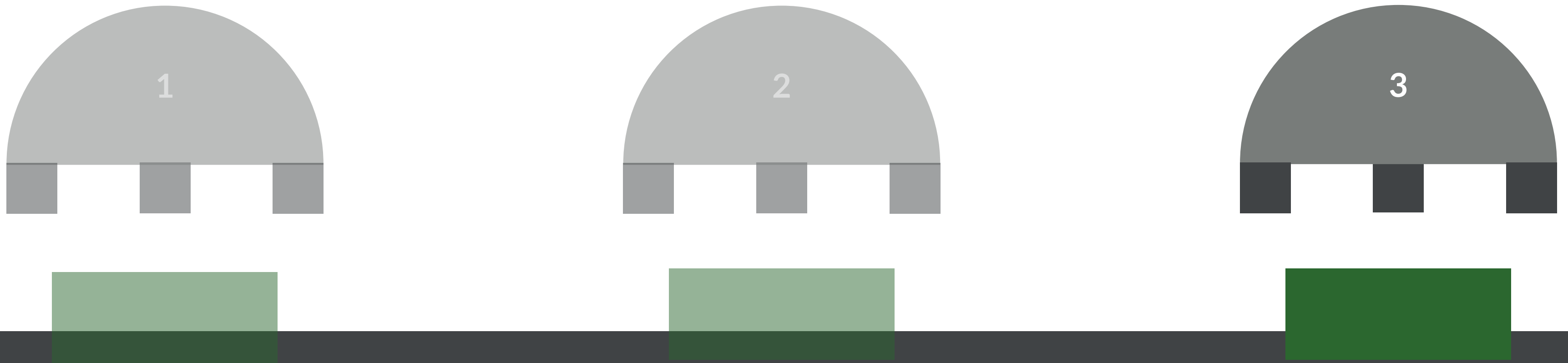


After machine 1 is finished, the conveyor belt moves the PCB along to machine 2. It picks up and places all the components it's tasked to on the PCB. This takes a certain amount of time.

Whilst this has happened a new PCB has rolled in under machine 1.

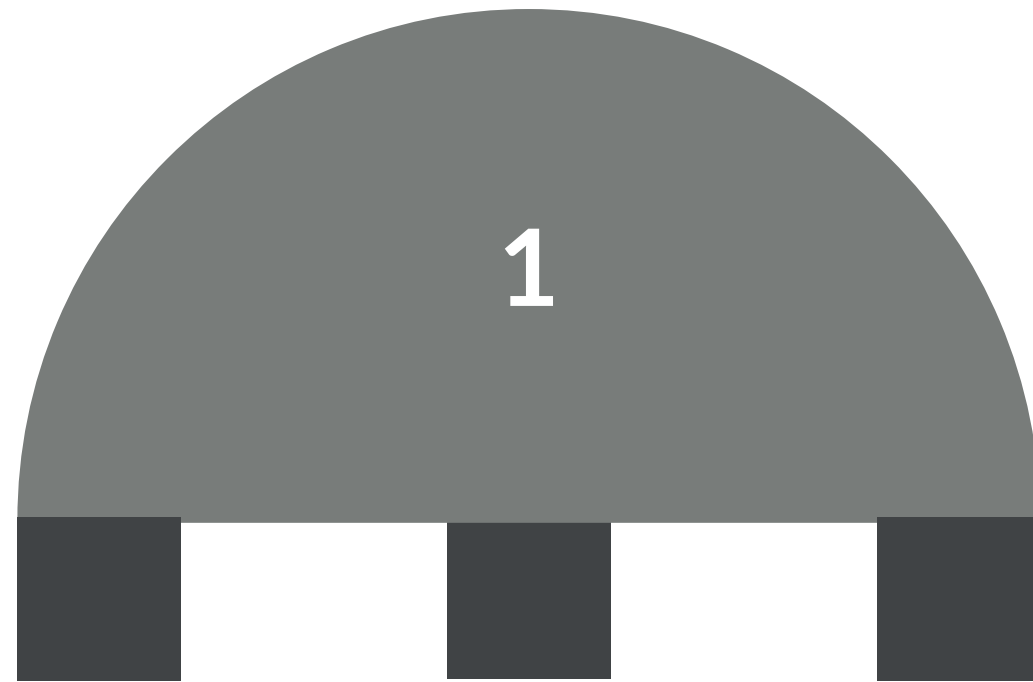


After either machine 1 or 2 is finished (whichever takes longest), the conveyor belt moves the PCB along to machine 3. It picks up and places all the components it is tasked to do. This takes a certain amount of time.



After machine 3 has finished, all the components required for the PCB **must have been placed**.

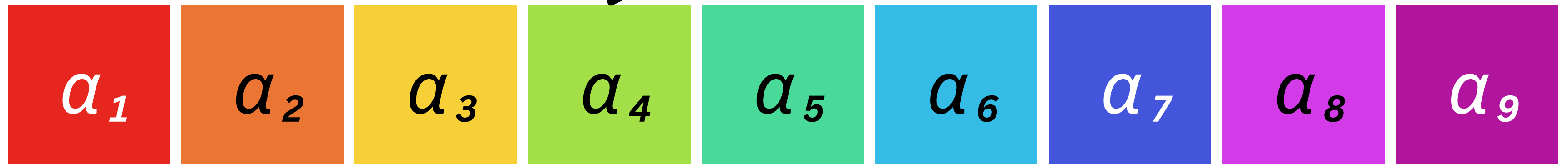
The conveyor belt moves at the speed of the slowest machine. If, for example, machine 1 takes 8 seconds, machine 2 takes 12 seconds, and machine 3 takes 4 seconds, then the conveyor belt will move every 12 seconds and it will take 36 seconds to make each PCB (after the first).

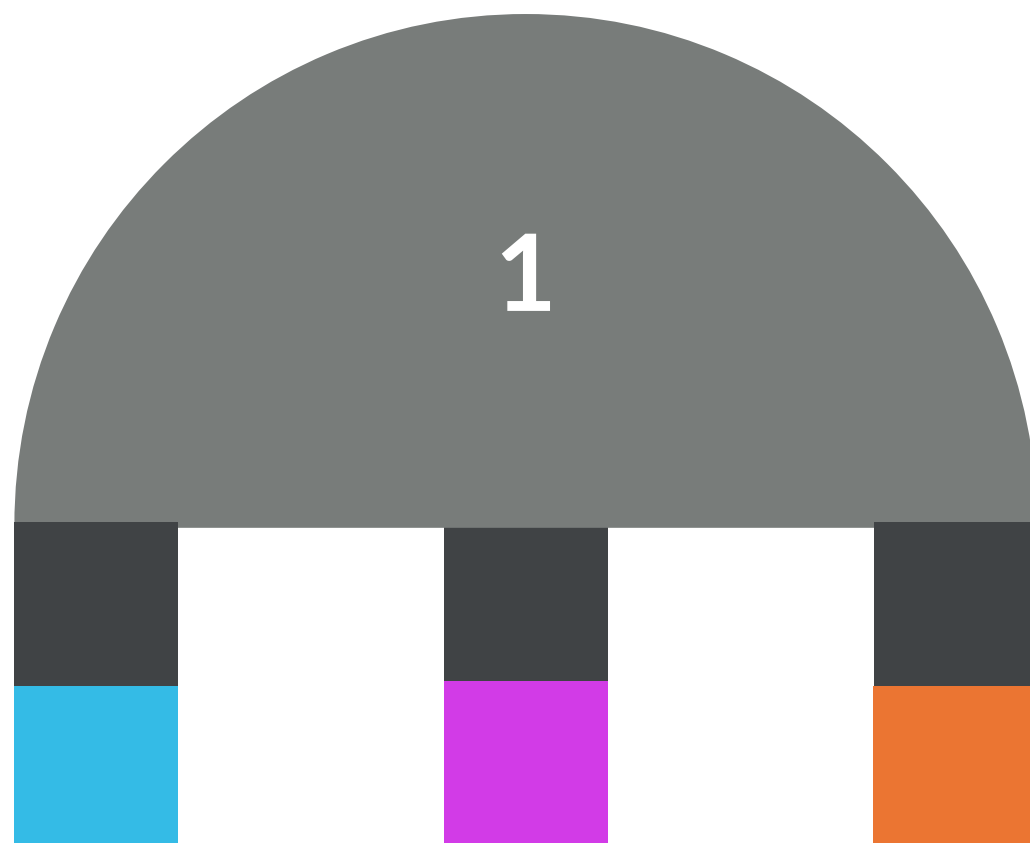


← Machine

← 3 Machine Attachment Heads

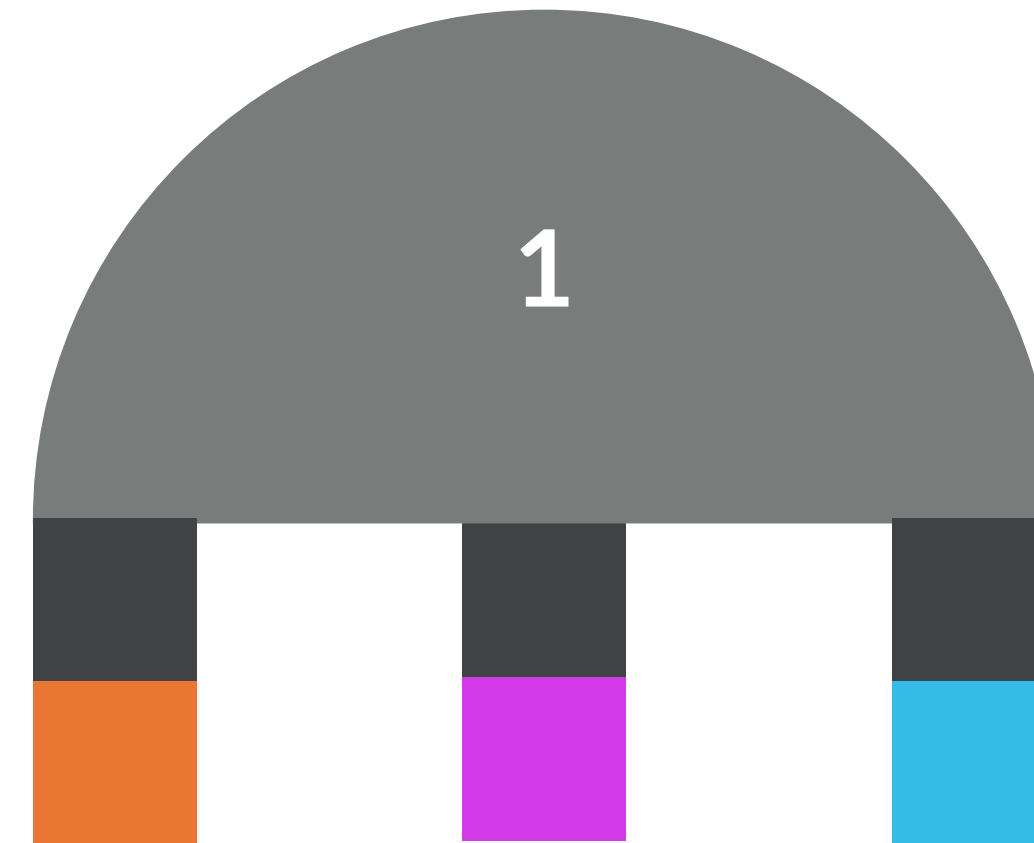
9 Attachments (only one of each).
Each attachment can only hold one
component at a time.





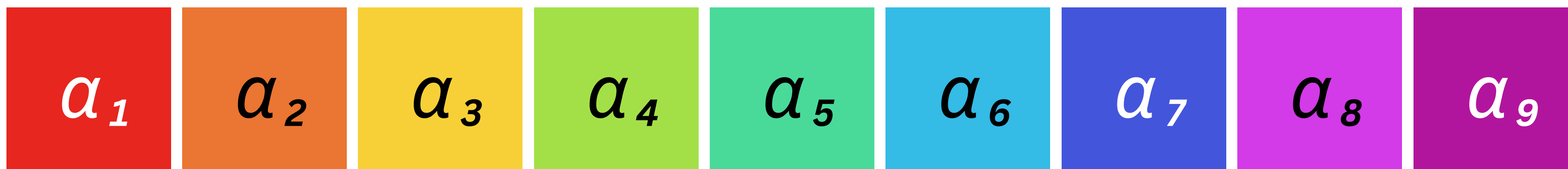
When picking up components from feeder bins, the 1st head picks first, then the 2nd, then the 3rd.

ATTACHMENT ORDER MATTERS

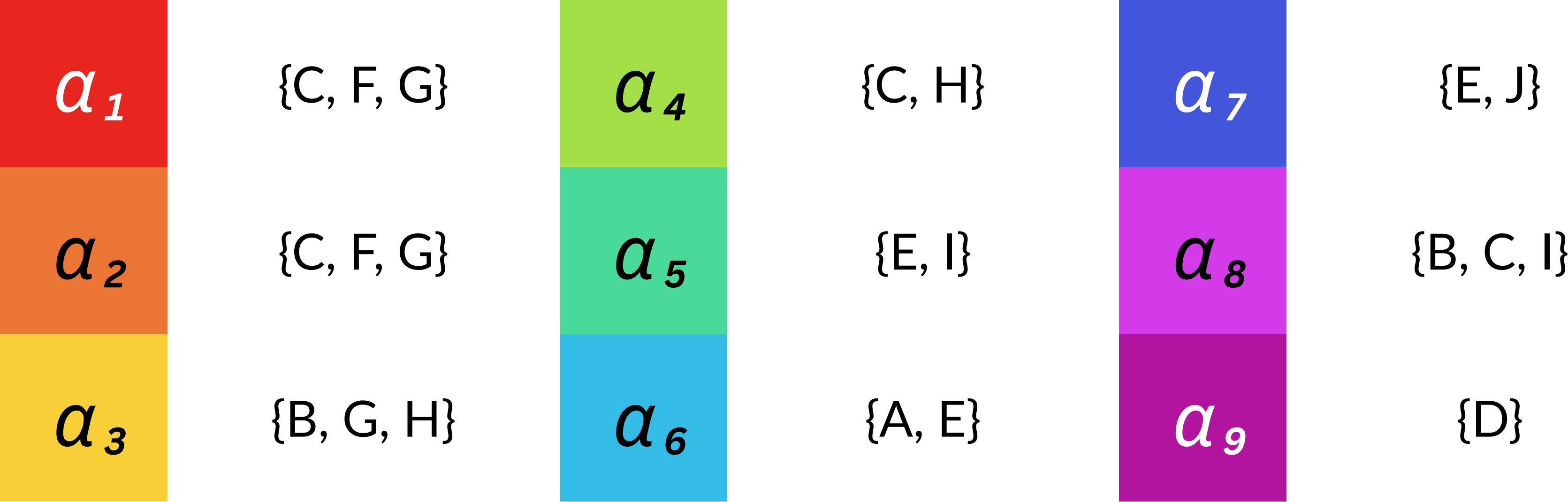


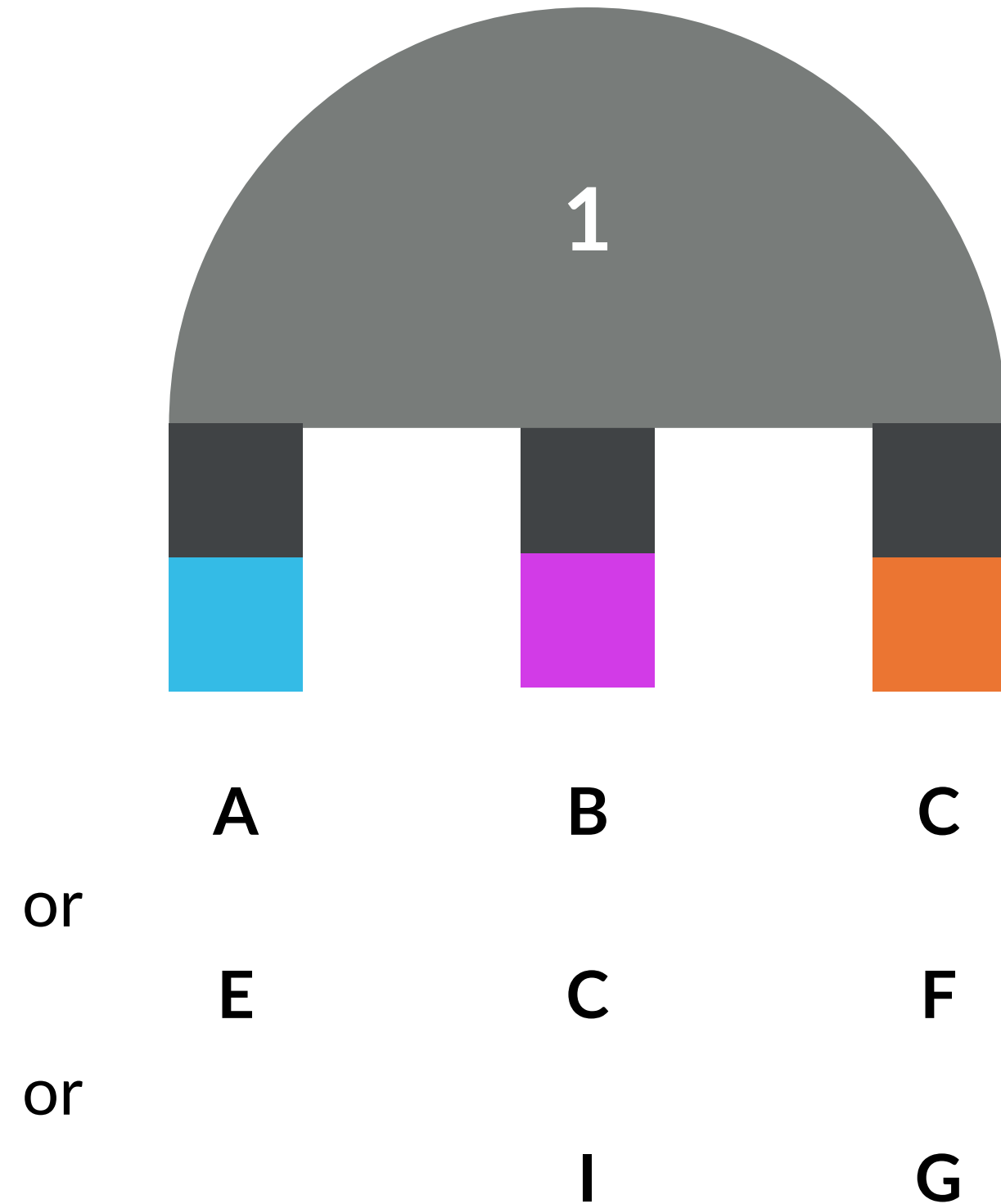
This will have different results...

...from this



Attachment-Component Map



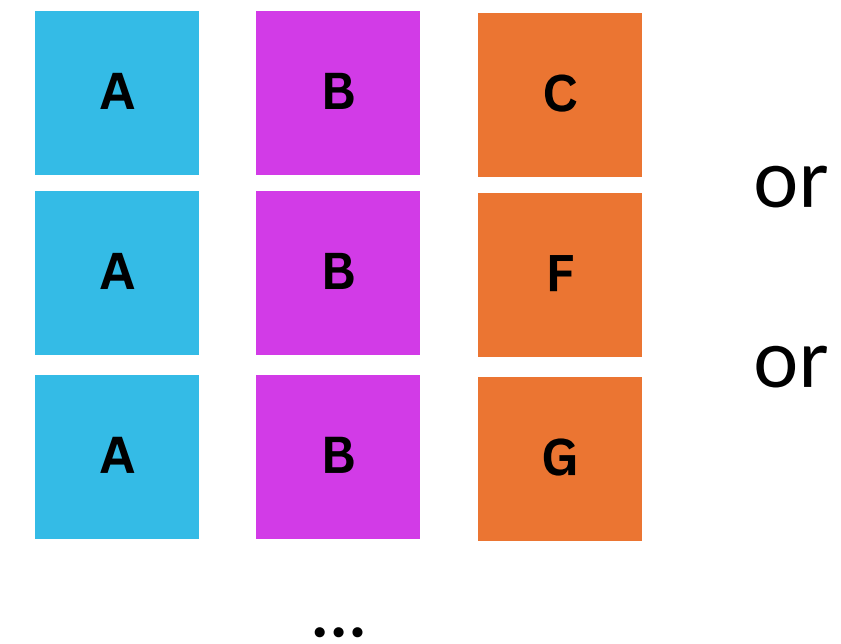


This attachment ordering means that if this machine picks up 3 components it will have to:

Travel to the **A** or **E** feeder bins first.

Then travel to **B**, **C**, or **I** feeder bins second.

Lastly, travel to the **C**, **F**, or **G** feeder bins third.



Pre-commencement

Each machine has an attachment fitted to its attachment heads. There is only one of each type of attachment. Once this has been decided there is no swapping.

Pick

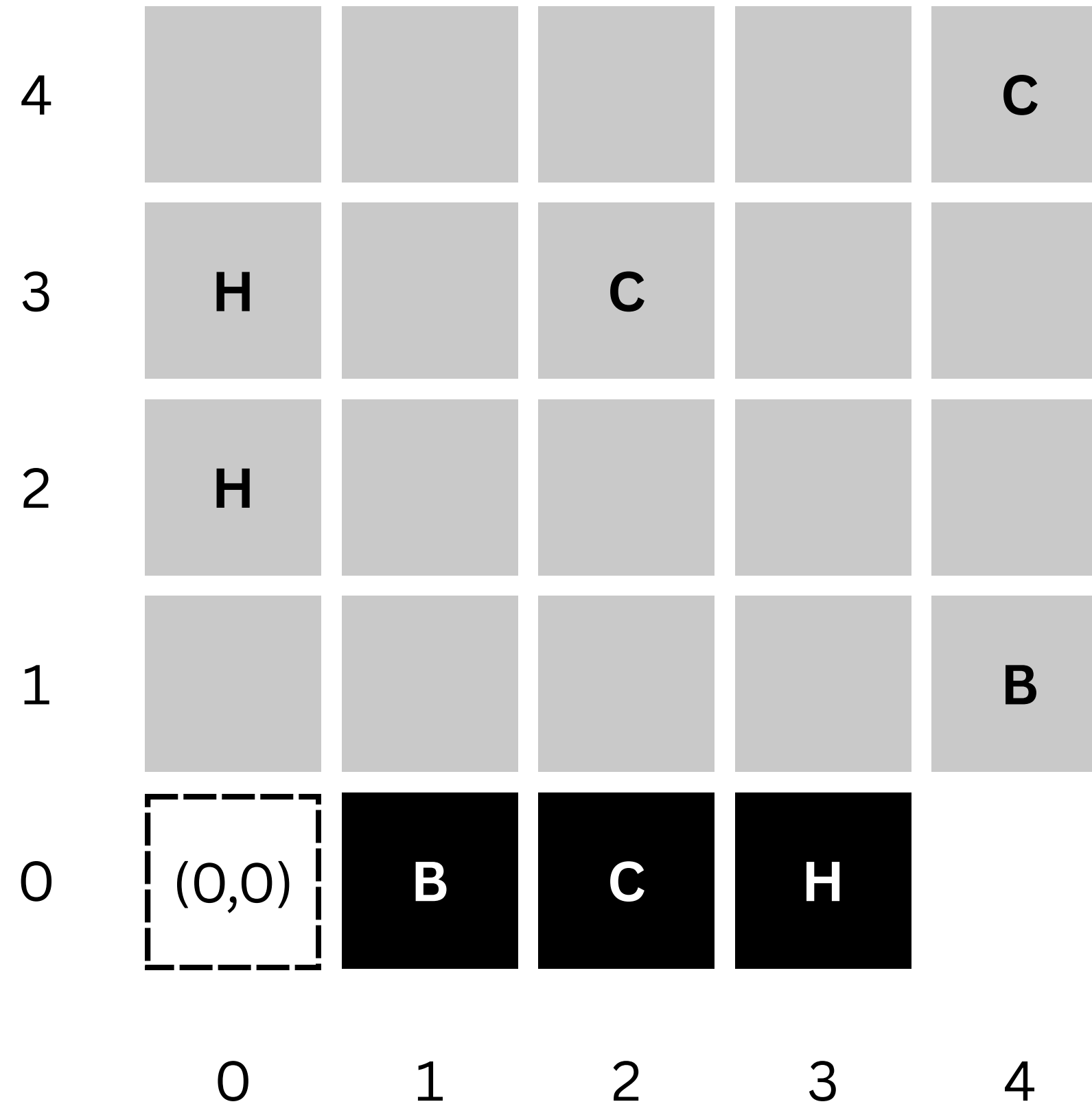
For each machine's pick and place phase, the machine picks up components from the feeder bins. This is done in order of the heads. The first head picks up its component, then the second, then the third. **This is why attachment order matters!** Each head can hold at most one component at a time.

Place

With the three components picked up, the machine can place them on the PCB board in any order it chooses.

Return

Having placed all components, the machine returns to the origin.



Minimal example (Single Machine)

This PCB (gray squares) requires 5 components (1xB, 2xC, 2xH).

The robot begins at the origin (0,0).

Each component can be received in head order from the feeder bins (black squares).

Component B must be placed at (4,1)

Component C must be placed at (2,3) and (4,4)

Component H must be placed at (0,2) and (0,3)

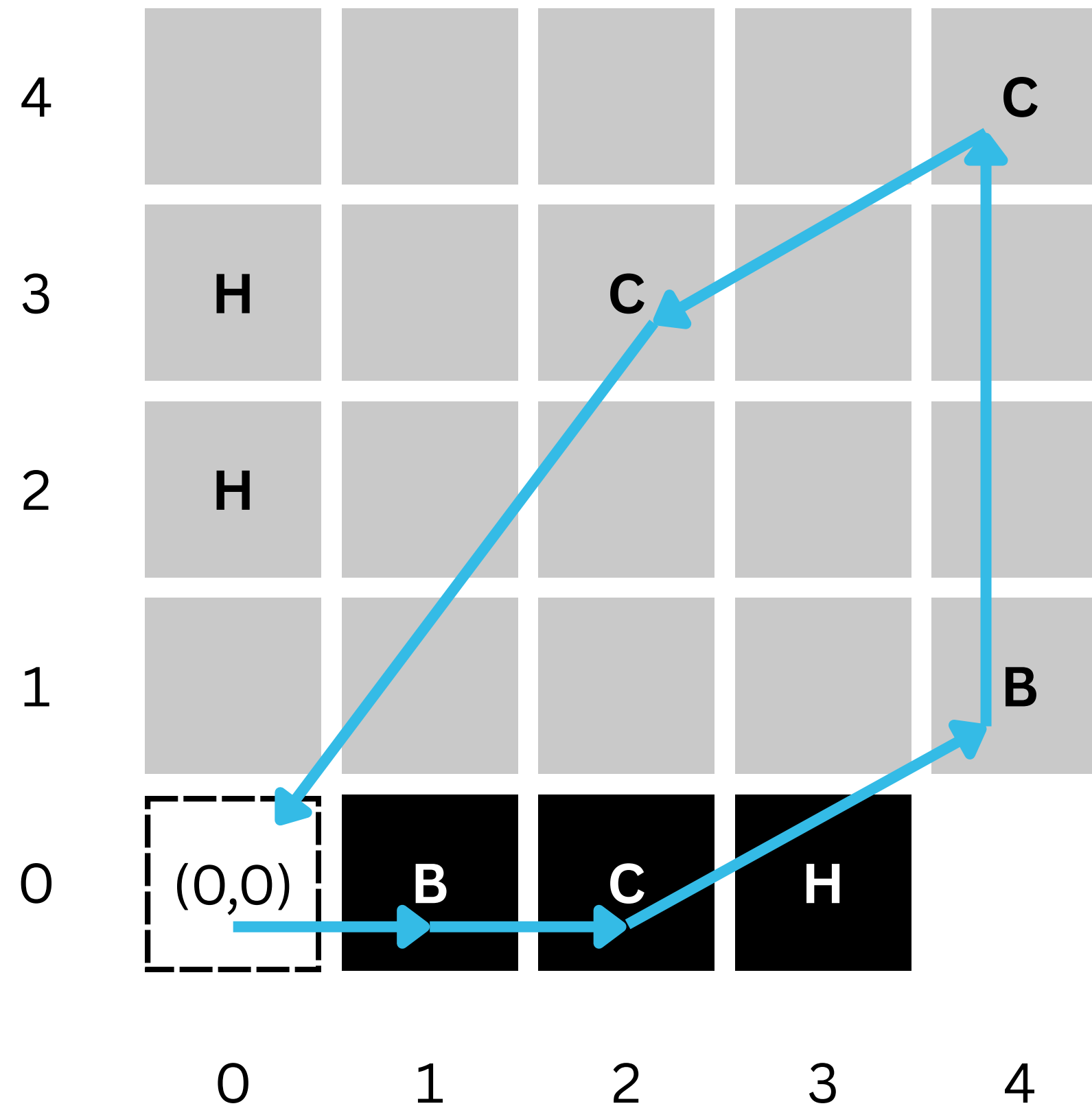
After each full placement the robot must return to the origin (0,0).

Minimal Example: Choosing Head Attachments

a_1	{C, F, G}	a_4	{C, H}	a_7	{E, J}
a_2	{C, F, G}	a_5	{E, I}	a_8	{B, C, I}
a_3	{B, G, H}	a_6	{A, E}	a_9	{D}

As we need attachments for components B, C, and H, we shall fit attachments 3, 4, and 8 in that order.





Pick and Place Phase 1

The machine begins at the origin (0,0).

It moves to (1,0) and picks component B onto head 1.

It moves to (2,0) and picks component C onto heads 2 and 3.

It moves to (4,1) and places component B.

It moves to (4,4) and places component C.

It moves to (2,3) and places component C.

It returns to the origin.

Pick and Place Phase 1: Movement Summary

Pick B onto Head 1

$(0,0) \rightarrow (1,0)$

$$d = \sqrt{(1-0)^2 + (0-0)^2} = 1$$

Pick C onto Heads 2 and 3

$(1,0) \rightarrow (2,0)$

$$d = \sqrt{(2-1)^2 + (0-0)^2} = 1$$

Place component B

$(2,0) \rightarrow (4,1)$

$$d = \sqrt{(4-2)^2 + (1-0)^2} = \sqrt{5}$$

Place component C

$(4,1) \rightarrow (4,4)$

$$d = \sqrt{(4-4)^2 + (4-1)^2} = 3$$

Place component C

$(4,4) \rightarrow (2,3)$

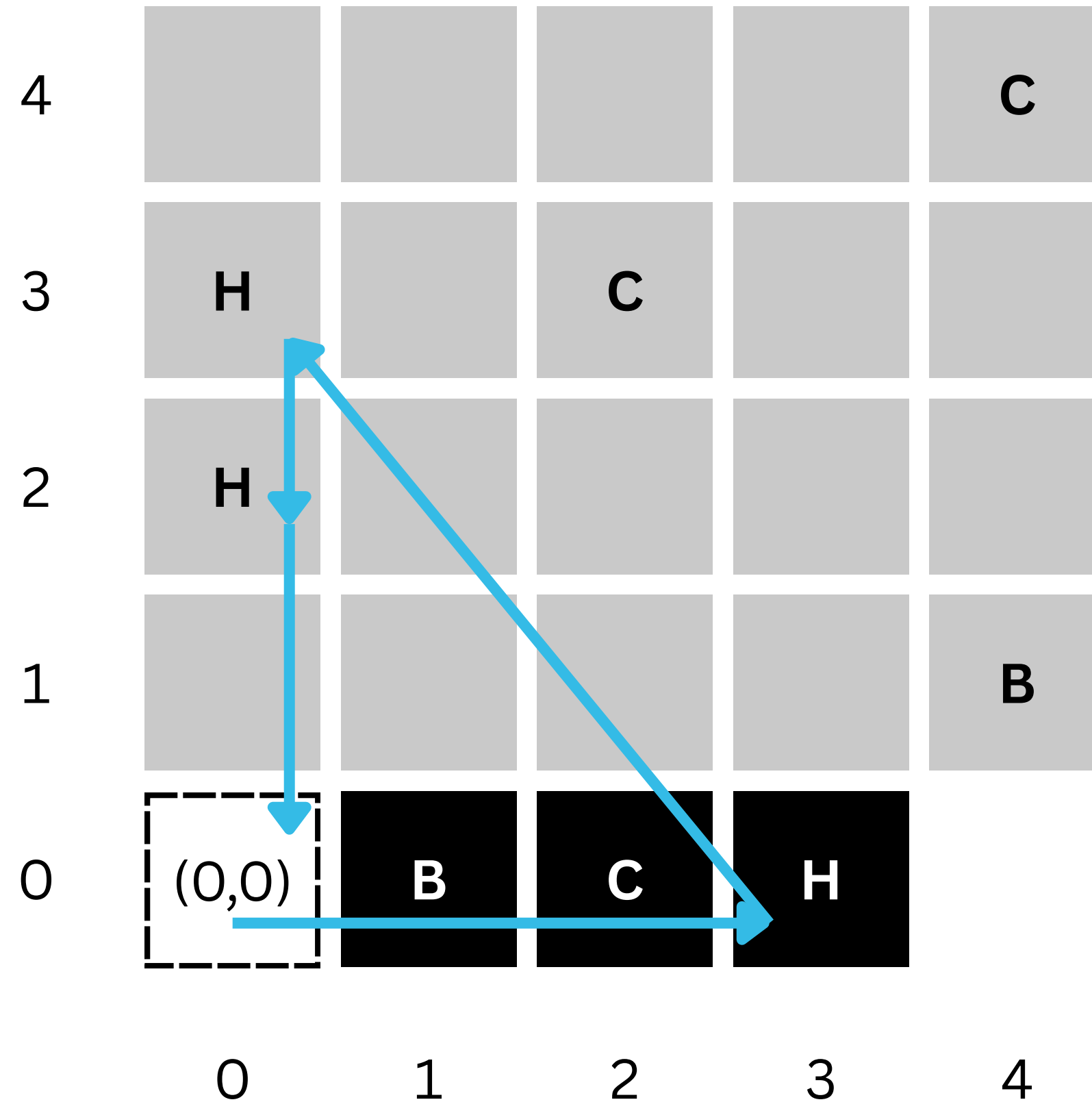
$$d = \sqrt{(2-4)^2 + (3-4)^2} = \sqrt{5}$$

Return to origin

$(2,3) \rightarrow (0,0)$

$$d = \sqrt{(0-2)^2 + (0-3)^2} = \sqrt{13}$$

Total movement: 13.08 (approx to 2 d.p.)



Pick and Place Phase 2

The machine begins at the origin (0,0).

It moves to (3,0) and picks component H onto heads 1 and 2.

It moves to (0,3) and places component H.
It moves to (0,2) and places component C.

It returns to the origin.

All components are now placed and the PCB is complete.

Pick and Place Phase 2: Movement Summary

Pick H onto Heads 1 and 2

$(0,0) \rightarrow (3,0)$

$$d = \sqrt{(3-0)^2 + (0-0)^2} = 3$$

Place component H

$(3,0) \rightarrow (0,3)$

$$d = \sqrt{(0-3)^2 + (3-0)^2} = \sqrt{18}$$

Place component H

$(0,3) \rightarrow (0,2)$

$$d = \sqrt{(0-0)^2 + (2-3)^2} = 1$$

Return to origin

$(0,2) \rightarrow (0,0)$

$$d = \sqrt{(0-0)^2 + (0-2)^2} = 2$$

Total movement: 10.24 (approx to 2 d.p.)

Total movement across all phases

(Phase 1) 13.08 + (Phase 2) 10.24 = 23.32



Can this be improved upon?

We assume that the time to pick up components from the feeders and placing them on PCB is negligible.

All the time is in travel distance.

Can we minimize the distance a machine has to travel to pick and place its components? Does a better solution exist?

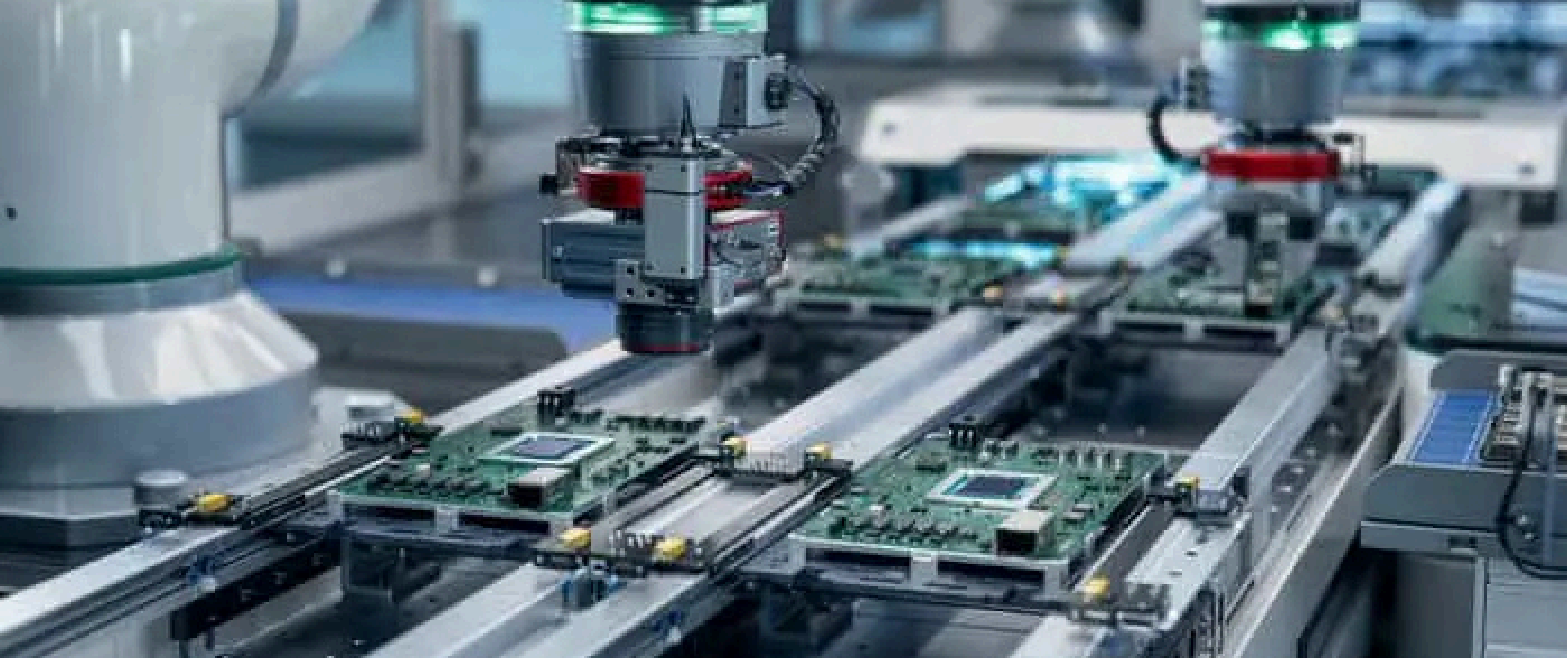
What your problem is

You are tasked with finding an attachment allocation for the three machines, and assigning pick and place tasks to the three machines such that:

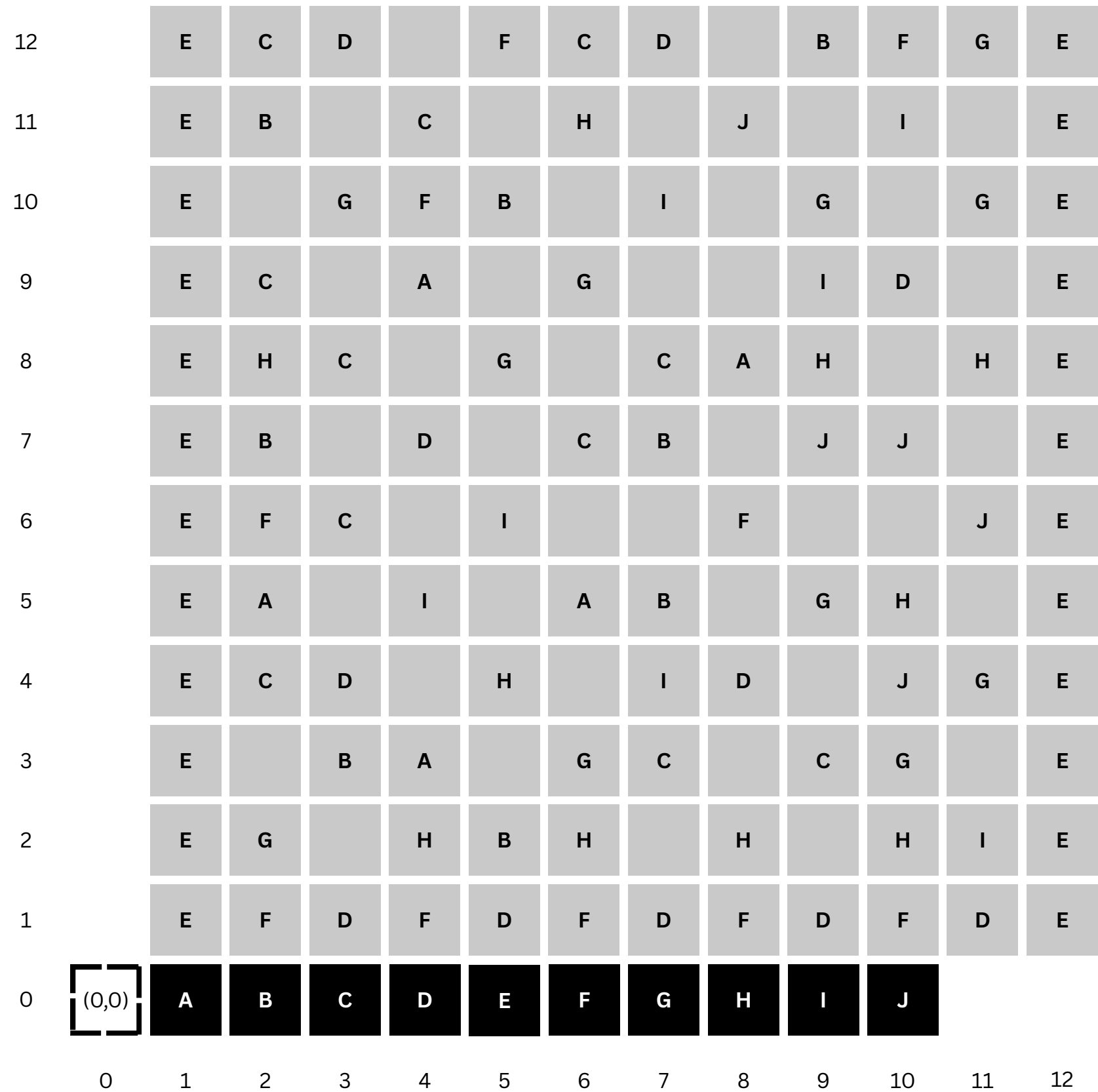
- 1) The PCB is completed as quickly as possible (minimum distance)
- 2) All components required by the PCB schematic are present and accurately placed.
- 3) The workload should be reasonably balanced across the three machines.

However, I don't want just a feasible solution

Your group is acting as a consultancy firm. It must solve the problem, report on it, summarize the solution found, explain the methodology taken, present findings to the board, and evidence complete team participation.



ASSIGNMENT PARTICULARS



PCB and Feeder Bin Locations

The PCB requires 102 total components.

A : 5

B : 8

C : 11

D : 11

E : 24

F : 10

G : 11

H : 10

I : 7

J : 5

An excel of **component locations** can be found on Blackboard.

Report (60%)

The report should be addressed to the manager of your consultancy firm and include details of the methodology you have employed, the results you have obtained, and comments on the way that your group has worked. The report should consist of a justification of the methods that you have used and a sufficiently detailed explanation to enable your manager to reproduce your methodology if needed.

Ensure that ALL of the student ID numbers of the group members are clearly listed on the front page.

Report (60%)

Distinction: 42–60 marks

The report briefly describes the problem, focussing on its technical characteristics. A logical approach to tackling the problem is described and justified. This is in sufficient detail for it to be repeatable by another team of consultants. The language used to describe the methodology is appropriate, and any mathematics used is correct, consistent and accompanied by adequate explanation. The group may have come up with a way of assessing how good their solution is and have some computer tools written for performing part, or all, of the optimisation process. An explanation of the recommended methodology for tackling problems of this nature is included in the report, and the results are given for applying the methodology to the particular problem instance that you have been given. Any literature referred to is explicitly referenced. The amount of detail included and the language used are appropriate for a consultancy manager.

Merit: 36–41 marks

The report includes most of the attributes of a distinction-level report, but descriptions of methods may not be completely clear, and/or the analysis of the problem may not be as thorough.

Pass: 30–35 marks

The report has some of the attributes of a distinction-level report. However, the methods and their descriptions contain minor mistakes.

Fail: <30 marks

The report has few of the attributes of a distinction-level report and may be seriously lacking in content and presentation

Solution (20%)

Your solution should include any source code, and input and output files used to evidence your solution. Appropriate technical documentation and comments detailing the solution are expected to be included. The format for the final output files can be found on Blackboard.

The score is calculated using an automated program which simulates the PCB assembly machine. The lower the score, the better. The program adds penalties based on the average performance of your solution whenever there are idling or waiting machines/heads since they could do some work but are not

Solution (20%)

Distinction: 14–20 marks

The score of the solution is below an efficient minimum, and there are no unplaced components or errors.

Merit: 12–13 marks

The solution is below a reasonable minimum, and there are no unplaced components and only minor errors.

Pass: 10–11 marks

The solution is below an acceptable minimum with only minor errors, or the solution looks to have the potential to be below a reasonable minimum, but there are some errors.

Fail: <10 marks

The solution is greater than an acceptable minimum , or there are too many errors to be calculated.

Solution Output

3 csv files, one for each machine. They should look like this...

x	y	component	action
0	0	-	origin
1	0	A	pick
4	3	A	place
...	...	...	...

Group Work (20%)

You will need to work together to devise an appropriate methodology and then delegate the work involved. You should keep minutes of all meetings using the form provided on Blackboard and submit these as an appendix to the group report.

It is hoped that the work will be divided equitably among group members. If you feel that a group member is not contributing to the work being carried out, please contact me.

Note that if some of your group have decided to work on the weekend and others are not able to, then please note this in your minutes, although no penalty will be applied for those unable to contribute over the weekend if the others in the group wish to continue working.

Group Work (20%)

Distinction: 14–20 marks

Meeting minutes and a copy of the ground rules have been submitted. They show effective group work, with all team members contributing and tasks appropriately assigned within the team. There is evidence of devising and implementing a clear plan for remote group work, with adaptations made where needed.

Merit: 12–13 marks

Minutes and ground rules have been submitted, but there is limited evidence of effective group work.

Pass: 10–11 marks

The report is missing minutes, or no ground rules have been submitted.

Fail: <10 marks

There is very little evidence of any group work.

A photograph of three male sprinters in their starting blocks on a red running track. The athlete on the left is wearing a black singlet with bib number 675 and is in lane 5. The athlete in the middle is wearing a blue singlet with bib number 622 and is in lane 4. The athlete on the right is wearing a white singlet with bib number 283 and is in lane 2. A fourth athlete is partially visible in lane 1. The track has white lane markings and numbers. A starting gun is visible in the foreground.

What we're going to do now...

- Setup and establish groups
- Look at groupwork agreement
- Go over any further points of clarification
- Group meets with James